

Blue light of frequency 6.25×10^{14} Hz is shone onto the sodium photocathode of a photocell. The graph of the photoelectric current versus potential difference is shown in Figure 3.

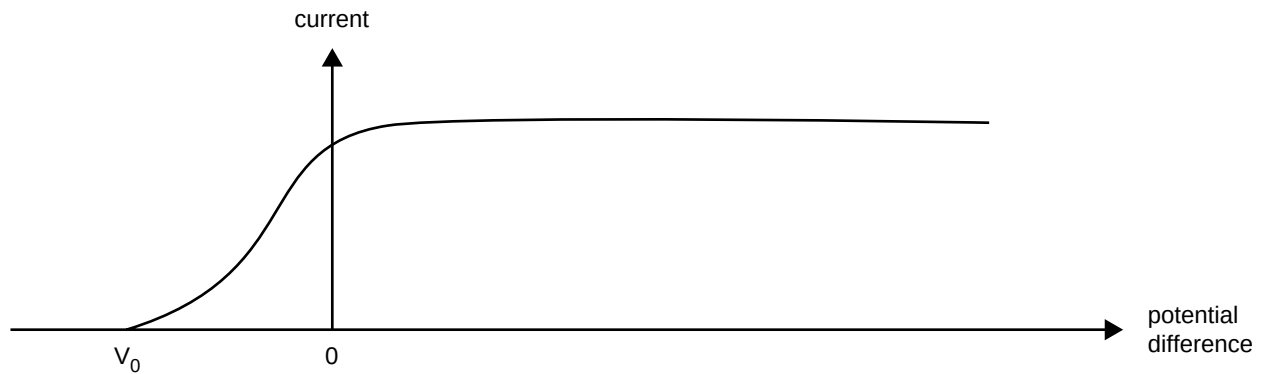


Figure 3

The threshold frequency for sodium is 5.50×10^{14} Hz.

Question 6

What is the cut-off potential, V_0 , when blue light of frequency 6.25×10^{14} Hz is shone onto the sodium photocathode of this photocell.

$$(h = 4.14 \times 10^{-15} \text{ eV s})$$

V

3 marks

Question 7

On **Figure 3** sketch the curve expected if the light is changed to **ultraviolet** with a **lower intensity** than the original.

2 marks

END OF QUESTION AND ANSWER BOOK

PHYSICS

Written examination 2

DATA SHEET

Directions to students

Detach this data sheet before commencing the examination.

This data sheet is provided for your reference.

1	velocity; acceleration	$v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$
2	equations for constant acceleration	$v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$
3	Newton's second law	$F = ma$
4	circular motion	$a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$
5	Hooke's law	$F = -kx$
6	elastic potential energy	$\frac{1}{2}kx^2$
7	gravitational potential energy near the surface of the Earth	mgh
8	kinetic energy	$\frac{1}{2}mv^2$
9	torque	$\tau = Fr$
10	Newton's law of universal gravitation	$F = G \frac{M_1 M_2}{r^2}$
11	gravitational field	$g = G \frac{M}{r^2}$
12	stress	$\sigma = \frac{F}{A}$
13	strain	$\epsilon = \frac{\Delta L}{L}$
14	Young's modulus	$E = \frac{\text{stress}}{\text{strain}}$
15	electric force on charged particle in an electric field	$F = qE$
16	electric field between charged plates	$E = \frac{V}{d}$
17	energy change of charged particle moving between charged plates	$\Delta E_k = qV$
18	photoelectric effect	$E_{k\max} = hf - W$
19	photon energy	hf
20	photon momentum	$p = \frac{h}{\lambda}$
21	de Broglie wavelength	$\lambda = \frac{h}{p}$

Gravitational field strength at the surface of Earth	$g = 9.8 \text{ N kg}^{-1}$
Universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$M_E = 5.98 \times 10^{24} \text{ kg}$
Radius of Earth	$R_E = 6.37 \times 10^6 \text{ m}$
Mass of the Sun	$M_{\text{SUN}} = 2.0 \times 10^{30} \text{ kg}$

Mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
Charge on the electron	$e = 1.6 \times 10^{-19} \text{ C}$
Planck's constant	$h = 6.63 \times 10^{-34} \text{ J s}$ $h = 4.14 \times 10^{-15} \text{ eV s}$
Speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$

Prefixes/Units

$$\text{m} = \text{milli} = 10^{-3}$$

$$\mu = \text{micro} = 10^{-6}$$

$$\text{n} = \text{nano} = 10^{-9}$$

$$\text{k} = \text{kilo} = 10^3$$

$$\text{M} = \text{mega} = 10^6$$

$$\text{G} = \text{giga} = 10^9$$

$$\text{tonne} = 10^3 \text{ kg}$$

END OF DATA SHEET